Big Data Analysis

Final Project Proposal

Does Applying Together Matter? Joint versus Single Mortgage Applications in the 2024 HMDA Data

Primary tools: Python, Pandas, PySpark | Dataset: HMDA 2024

Instructor: Huan Vu, Faculty of DS&AI
Group members: Trần Thu Thuỷ, Nguyễn Gia Phong, Nguyễn Tuấn Sỹ

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1 Problem and Motivation

A household applying for a mortgage can apply alone or with a co-applicant, such as a spouse or partner. A co-applicant can add a second income, which may make the loan look more affordable, but joint applicants may also request larger loans. Whether this helps or hurts the lender's decision is not obvious, so we measure it. The answer matters to lenders (the mix of applications they receive), to analysts and policymakers (access to housing credit), and to borrowers (whether applying jointly is linked to a different outcome). We use 2024 HMDA data because it records every reported application, including denied and withdrawn ones. All findings are associations; we make no causal claim and do not treat any gap as proof of discrimination.

2 Research Questions

Main question. How do mortgage application outcomes and loan amounts differ between applications with a co-applicant (joint) and without one (single) in 2024?

RQ1. How does the denial rate differ between joint and single applications?

RQ2. How do income, loan amount, and loan-to-income ratio differ between the groups?

RQ3. Does the denial-rate gap remain among applications with similar income, debt-to-income ratio, loan type, and loan purpose?

RQ4. Does the gap vary by loan purpose and across the five states with the most applications?

3 Data and Key Definitions

We use the instructor-provided 2024 HMDA loan-level file (CFPB/FFIEC). One row is one reported application, not necessarily an issued loan. We will first record the file's format, size, number of records and variables, and verify every variable against the official 2024 data dictionary. Key definitions:

- **Joint indicator:** `joint_flag` $= 1$ if the co-applicant fields show a co-applicant, 0 if they show none.
- **Denial rate:** $\dfrac{D}{O + A + D}$, where O = originated, A = approved but not accepted, D = denied. Only applications with a lender decision count; withdrawn, incomplete, purchased-loan, and preapproval-only records are excluded and reported separately.
- **DTI:** debt-to-income ratio, reported mostly in ranges, so used as ordered bands. **LTI:** loan amount divided by income, computed only for valid positive values.

4 Methods

Data preparation. Special codes such as "Exempt" are treated as missing, never as zero. We check duplicates, keep plausible extreme values (large loans can be real), and remove only impossible ones such as negative amounts. All rules are documented and applied identically in both tools; cleaned data is saved as Parquet, a compressed format both tools read quickly.

Analysis. We report outcome shares and medians with interquartile ranges (more reliable than means for skewed income and loan amounts), then compare groups by loan type, purpose, and state. For RQ3 we compare denial rates within cells of income quartile × DTI band × loan type × purpose and average the gaps, weighted by cell size. If the average gap is much smaller than the raw gap, applicant profile explains most of it. We judge results by size in percentage points, since millions of records make almost any difference statistically significant. Visuals: a bar chart of denial rates, box plots of loan amount, and a heatmap by income and DTI.

Pandas and PySpark. Pandas analyzes data in one computer's memory; PySpark processes it in parallel partitions. Both run the same workflow on identical records. We compare at least three outputs: record counts after cleaning, denial rates by `joint_flag`, and mean and median loan amount by loan type. Counts must match exactly and decimals within a small tolerance; any discrepancy is investigated, not hidden.

Performance. The workload groups applications by `joint_flag`, loan type, and state and computes counts, denial rate, and loan-amount statistics, on 25%, 50%, and 100% of the data. Only the analysis is timed (loading and Spark startup are separate); Spark runs end with an action because it evaluates lazily. We repeat five times, report the median, and compute speedup as $T_{\text{Pandas}}/T_{\text{Spark}}$, with the hardware recorded. We do not assume Spark is faster.

5 Optional Predictive Extension

Only after all required parts are done: predict denied vs. originated using information known at application (joint indicator, loan amount, income, LTI, DTI, loan type, purpose, state). Denial reasons, interest rate, and loan costs are excluded to avoid data leakage. We use logistic regression, then a random forest, with a stratified 80/20 split, and report precision, recall, F1, ROC-AUC, and PR-AUC against a majority-class baseline.

6 Limitations and Team

A co-applicant is not marital status; income in joint applications may be combined; credit history is not available in the public file; and one year shows no trend. Gaps are patterns to describe, not proof of discrimination or cause.

Member	Main responsibility
Trần Thu Thuỷ	Data inspection, cleaning, features; report writing
Nguyễn Gia Phong	Pandas implementation, descriptive analysis, visualizations
Nguyễn Tuấn Sỹ	PySpark implementation, equivalence checks, performance experiment

Every member reviews all work. Presentation: 16 November 2026; final report one week later.

Sources: `https://ffiec.cfpb.gov/data-publication/`, `https://ffiec.cfpb.gov/documentation/`.